Beyond Structural Fidelity: A Function-Aware Benchmark of Backbone Extraction for Spreading and Community Preservation in Weighted Networks

Oct 18, 2026 · @Tanaz

1. Project at a glance

The question. A backbone keeps a small share of a weighted network's edges. It is usually judged by structural checks such as connectivity and the weight and degree distributions. This project asks when a backbone that passes those checks still fails to behave like the original network, and why.

Central contribution: a structure-versus-function failure analysis. The project (1) measures how well structural fidelity predicts functional fidelity, meaning preservation of spreading, communities, shortest paths and node rankings, across six methods, eight edge budgets and many networks; (2) maps where that prediction fails and tests candidate mechanisms on controlled synthetic networks; and (3) turns the findings into an advisor that recommends a method for a given network, objective and edge budget, with evidence and uncertainty. Benchmarking the six methods is the instrument, not the claim.

Contribution statement (for your faculty): We identify when and why structural fidelity fails to predict functional fidelity in backbones of weighted networks, test the proposed mechanisms on controlled synthetic networks, and provide an advisor that recommends a backbone method with uncertainty-aware evidence.

What gets built

- Six backbone filters, written from scratch and cut to the same edge budgets.
- An evaluation suite with defined fidelity scores, random baselines and confidence intervals.
- A failure analysis: agreement tests, a failure atlas and mechanism tests.
- A Backbone Advisor with an evidence mode, and a predictive mode only if it generalizes to unseen networks.
- An interactive dashboard that shows all of this.

The demo in one line. Move the edge budget, watch six methods shrink the same network, see which backbones still spread, cluster and route like the original, find the cases where structural checks said fine but function broke, and read the advisor's recommendation with its evidence.

Honest status. This is a strong direction, not yet a proven novel paper. Earlier work already covers parts of the question (section 3), so the claims rest on the failure analysis and its mechanism tests, and they apply only to the networks and dynamics actually tested.

2. Problem statement

What a backbone is. Many real networks are weighted and dense, such as airports linked by passenger counts or people linked by time spent face to face. Backbone extraction keeps a subset of edges that carries the important structure. The disparity filter of Serrano et al. keeps edges that are locally significant against a null model for how weight is assigned. The NetBone paper lists the benefits: less data, faster graph algorithms, interactive analysis and noise removal.

The problem. Methods differ in what they call important: some test edge weights statistically, some keep shortest paths, some use spectral properties. A method is usually chosen by structural checks, such as whether all nodes stay connected or whether the weight and degree distributions look similar. But backbones are used for tasks: simulating an epidemic on a smaller network, finding communities, routing, ranking nodes. Structural similarity does not guarantee behavioural similarity. Yan et al. (2018) found that conventional network properties are disrupted early by weight thresholding and that many standard measures behave unreliably under edge removal, and Pereira et al. (2026) report that weight thresholding and the disparity filter remove bridges early and hinder spreading. Nobody I found has measured, across methods, budgets and networks, how often and why structural fidelity fails to predict functional fidelity.

Problem statements

1. *When does a backbone that preserves structure fail to preserve function (spreading, communities, shortest paths, node rankings), and what explains the failures?*
2. *Given a weighted network, an objective and an edge budget, which method should be used, and how confident is that recommendation?*

Scope. Undirected, static, weighted networks of up to a few hundred nodes and tens of thousands of edges for the main experiments, so everything runs on a CPU. Contact data are aggregated over time. Dynamics are SI and SIR with the weight-to-probability map defined in section 6. Directed, bipartite and temporal networks are out of scope.

3. Related work and novelty assessment

The base idea comes from Serrano et al. (2009) and the methods that followed it. The first table lists what each paper established and what that means for this project. The second states which claims the project can and cannot make as a result.

Paper	What it shows	Consequence for this project
Serrano, Boguñá & Vespignani (2009)	The disparity filter keeps edges that are locally significant against a null model	Base paper; method 2
Coscia & Neffke (2017)	The noise-corrected filter estimates noise in edge weights and keeps edges that differ enough from a random benchmark	Method 3
Grady, Thiemann & Brockmann (2012)	The high-salience skeleton; the authors report that salience predicts features of contagion	Method 5
Simas, Correia & Rocha (2021)	The distance (metric) backbone keeps the edges that lie on shortest paths	Basis of method 4

Paper	What it shows	Consequence for this project
Hamann et al. (2016)	Describes itself as the first systematic comparison of edge sparsification methods on diverse properties: connectivity, communities, centralities and epidemic simulations, on social networks. Different methods are best for different properties	A multi-property comparison already exists, so comparing methods is not a contribution here
Yan et al. (2018)	For weight thresholding, conventional network properties are disrupted early and many standard measures behave unreliably under progressive edge removal	Closest hint that structural measures mislead, limited to thresholding
Yassin et al. (2023)	NetBone: 20 methods and a comparison framework built on structural properties	Reference implementation and the usual structural way of choosing a method
Statistical filter comparison (2024)	Compares seven statistical filters across diverse networks	A plain filter comparison is done
Correia, Barrat & Rocha (2023), PLoS Comput. Biol. 19(2), e1010854	Contact networks have small metric backbones that maintain community structure and are primary transmission subgraphs	Metric backbones preserving communities and spreading is already known
Soriano-Paños, Costa & Rocha (2025)	Semi-metric distortion sparsification recovers spreading better than weight and effective-resistance thresholding, on synthetic and real networks	Method 4; supports H4

Paper	What it shows	Consequence for this project
Mercier, Scarpino & Moore (2022)	Effective-resistance sparsification preserves SIR behaviour on a mobility network	Method 6
Pereira, Costa & Rocha (2026, preprint)	On 12 contact networks, shortest-path-redundancy methods preserve eigenvector-centrality ranks and infection timing better than weight thresholding, the disparity filter and weighted effective resistance; weight thresholding and the disparity filter remove bridges early and disconnect the network	The bridge explanation is already proposed for spreading, so the mechanism work must go further
Kirkley (2025)	MDL-based backbones; compares six methods on 69 real weighted networks for edges kept, connectivity and strength distributions	Source of structural metrics and of a network corpus

Novelty assessment

Candidate claim	Status in the sources I read	Decision
Comparing many methods on many properties at matched edge budgets	Done by Hamann et al. and by several backbone comparisons	Not a contribution; the benchmark is the instrument
Distance and metric backbones preserve spreading and communities	Shown by Correia et al., Soriano-Paños et al. and Pereira et al.	Not a contribution; replicated as H4

Candidate claim	Status in the sources I read	Decision
Weight thresholding and the disparity filter hurt spreading by removing bridges	Reported for contact networks by Pereira et al.	Replicate, extend to communities and paths, and test the mechanism by controlled manipulation
Structural fidelity does not predict functional fidelity	Hints in Yan et al. (thresholding only) and Hamann et al. (different methods suit different properties); I found no direct quantitative test across methods	Central question; confirm in week 1
A mechanism for the failures, validated on controlled synthetic networks	Not found	Core of the discovery
An evidence-based method advisor with a generalization test	Not found	Secondary contribution; predictive mode is conditional

Claims to avoid

- That this is the first comparison of sparsification methods on several properties.
- That spreading preservation under sparsification is untested.
- That the metric backbone, or any single method, is best. That depends on the goal and the network and is what the experiments test.
- That bridge removal explains the failures, before experiment E3 supports it.

How this was checked. I read the full texts of the NetBone, Kirkley and Pereira et al. papers and abstracts or summaries of the rest. In week 1, read Hamann et al., Yan et al., Correia et al. and Soriano-Paños et al. in full and repeat the search for a direct test of structure-to-function agreement. If one exists, narrow the central question to the part that remains.

4. Research questions and hypotheses

Terms. For a network G and a backbone B that keeps a share q of the edges, structural fidelity F_struct and functional fidelity F_g (for each function g: spreading, communities, paths, rankings) are scores from 0 to 1, where 1 means B behaves exactly like G. Section 6 defines them. A **cell** is one network at one budget. A **fidelity failure** is a case where the backbone passes the structural check but fails the functional one: F_struct at or above a threshold tau_s while F_g is below tau_f. The defaults tau_s = 0.8 and tau_f = 0.6 are set on a pilot set of synthetic networks before any real network is analysed.

Research questions

1. When does structural fidelity fail to predict functional fidelity? This covers both agreement and failure rate.
2. Why? Which properties of the method, the budget and the network explain the failures?
3. Does the answer depend on the function: spreading, communities, shortest paths or node rankings?
4. Which methods are robust, meaning they rarely fail, and where do filters beat random removal?
5. Conditional: can an advisor recommend a method that generalizes to networks it has not seen?

Hypotheses (written down before any experiment runs). Each states a direction and a test. The thresholds are defaults to be fixed in week 1.

- **H1, agreement is weak.** Within a cell, the ranking of the six methods by F_struct agrees only weakly with their ranking by F_g. Test: Kendall's tau-b per cell, averaged over cells, with a 95% cluster-bootstrap interval over networks. Supported if the mean and the upper limit of the interval are below 0.6 for at least two of the four functions.
- **H2, bridge mechanism.** Failures in spreading and communities occur where a method keeps proportionally fewer inter-community (bridging) edges than intra-community edges, measured by a bridge-retention ratio below 1. Test: mixed-effects regression of the gap F_struct minus F_g on the ratio, budget and method, with network as a random effect. Supported if the ratio has the predicted sign, survives Holm correction and improves the model over method and budget alone (likelihood-ratio test).
- **H3, weight heterogeneity.** On synthetic networks, more weight heterogeneity or stronger weight-degree coupling raises the failure rate of weight-based filters (global threshold, disparity filter) more than that of the shortest-path-redundancy method

(SMDS). Test: the interaction of method with each factor in the factorial experiment. Supported if the interaction interval excludes zero in the predicted direction.

- **H4, path-redundancy methods.** At budgets above the size of the metric backbone, SMDS has a smaller structure-to-function gap and a higher skill score on spreading and paths than the global threshold and the disparity filter. This replicates earlier reports on more networks. Test: paired Wilcoxon signed-rank on network-by-budget cells, Holm-corrected.
- **H5, function dependence.** No single method wins on all four functions: in more than half of the cells, at least two functions have different best methods. Test: the share of such cells with a cluster-bootstrap interval. Supported if the lower limit exceeds 0.5.

Decision rules. Each hypothesis is judged supported, rejected or inconclusive by its stated rule, with Holm correction across H1 to H5 at a family-wise level of 0.05. A rejected hypothesis is a finding. If H1 is rejected, structure predicts function well, and the advisor can rely on cheap structural proxies.

5. Methods compared and the matched-budget protocol

Six methods cover the main mechanism families, and every one of them ranks edges, so every one can be cut to any edge budget. Implement all six yourselves.

#	Method	Family	Edge score (the highest-ranked edges are kept)	Source
1	Global weight threshold (W)	Global, weight only	The weight w_ij	Baseline, as in Yan et al. (2018)
2	Disparity filter (DF)	Local, statistical	For each endpoint, the null probability (1 - p)^(k - 1) of a normalised weight p at least this large; the smaller of the two is kept as the significance	Serrano et al. (2009)

#	Method	Family	Edge score (the highest-ranked edges are kept)	Source
3	Noise-corrected filter (NC)	Local, statistical	How many posterior standard deviations the weight lies above its binomial expectation	Coscia & Neffke (2017)
4	Semi-metric distortion sparsification (SMDS)	Shortest-path redundancy	Distortion s = d_ij divided by the shortest indirect distance; smallest distortion first, so the metric backbone (s = 1) comes first	Soriano-Paños et al. (2025), Simas et al. (2021)
5	High-salience skeleton (HSS)	Shortest-path trees, global	Salience: the share of shortest-path trees that contain the edge	Grady et al. (2012)
6	Weighted effective resistance (wEffRes)	Spectral	Proximity times the effective resistance of the edge	Mercier et al. (2022)

Null baselines (not counted among the six): R-U removes edges uniformly at random; R-W keeps k edges sampled without replacement with probability proportional to weight.

Why these six. Each represents one mechanism with prior literature. The set includes the methods that Pereira et al. (2026) compare (W, DF, wEffRes, SMDS), which allows results to be checked against theirs, and adds a second local statistical filter (NC) and a path-salience filter (HSS). Left out of the core set: LANS, the maximum spanning tree, the doubly stochastic filter, the h-backbone and the Polya filter, which are either fixed-size or add little to the mechanism contrast, and the Distance Backbone Synthesis, whose per-edge parameter search is expensive. Any of them can be a stretch addition.

Weights, proximity and distance. Several methods need a path length. Use the convention of the contact-network literature: proximity $p_{ij} = w_{ij} / (s_i + s_j - w_{ij})$, where s_i is the strength of node i, which lies in [0, 1]; distance $d_{ij} = 1 / p_{ij} - 1$. Methods 1 to 3 use the raw weight. E5 tests the alternative $p_{ij} = w_{ij} / w_{max}$.

The matched-budget protocol

1. The budget q is the share of edges kept, and k = round(q times the number of edges). The grid is q = 0.05, 0.10, 0.15, 0.20, 0.30, 0.40, 0.50 and 0.70.
2. Each method returns a total order of the edges, and the backbone is the top k. All six core methods are score-based, so none returns a fixed-size backbone.
3. Ties are broken by weight (higher first), then by a seeded random permutation shared across methods. Report the share of tied edges at each budget.
4. The node set is fixed. A node that loses all its edges stays in the network as an isolate, so its effect is measured instead of hidden.
5. **Fixed-size structures.** The metric backbone is the first set of edges in the SMDS order (all with distortion 1) and serves as a built-in anchor. For budgets below its size, metric edges are ordered by weight and the budget is flagged as below the structural floor. The maximum spanning tree (n - 1 edges) is reported as a reference for the connectivity floor, not as a method. If a stretch method returns only one fixed subgraph, evaluate it at its native size, or extend it by adding the highest-scoring remaining edges, and label the result as extended.
6. Retained edges keep their original weights, with no reweighting. The effective-resistance method is the deterministic top-k variant. The randomised Spielman-Srivastava sampler reweights edges and is out of scope.
7. Report each method's connectivity floor: the smallest budget at which its backbone is connected. Metrics at disconnected budgets are kept and flagged.
8. **Validation before use.** DF, NC and HSS outputs must agree with the NetBone package on the Les Miserables network, with an agreement threshold fixed in week 1 (for example Jaccard similarity of the retained edge sets at equal budgets). SMDS must reproduce the metric backbone computed independently, and that backbone must preserve every shortest-path distance. wEffRes is checked against a hand-computed pseudoinverse on a small graph. NetBone does not implement SMDS or wEffRes, so those two rely on these independent checks.

6. Evaluation metrics, baselines and uncertainty

Every fidelity score compares the backbone B with the original network G on the same node set V, takes values from 0 to 1, and equals 1 when B behaves exactly like G. Notation: n is the number of nodes, s_i the strength of node i, W the sum of all strengths, and p_ij and d_ij the proximity and distance defined in section 5.

Function	Score	Definition	Noise source	Sanity check
Structure	F_struct	Mean of reachability ratio, non-isolated node share and one minus the Hellinger distance of strength distributions	None except ties	Equals 1 at q = 1
Spreading	F_spread	Mean of F_order (infection-order correlation) and F_time (half-infection-time ratio)	SI runs and seeds	Equals 1 at q = 1
Communities	F_comm	NMI between Louvain partitions of G and B, divided by Louvain's self-agreement on G	Louvain seeds	Equals 1 at q = 1
Paths	F_path	Share of connected node pairs whose shortest distance grows by at most a factor 1 + epsilon	None except ties	Equals 1 for SMDS at the metric-backbone size
Rankings	F_rank	Mean over three centralities of the Spearman correlation, clipped at 0	None except ties	Equals 1 at q = 1

Structure. F_struct is the mean of three components, each also reported alone. The reachability ratio is the share of ordered node pairs connected in B divided by the share connected in G. The non-isolated share is the fraction of nodes with at least one edge in B. The third component compares strength distributions after normalising by total weight, which removes the penalty that plain distribution tests (such as a KS test on raw degrees) impose on any sparser network. It is the form used by Kirkley (2025):

```
F_{\mathrm{str}} = 1 - \sqrt{\tfrac{1}{2}\sum_{i \in V}\left(\sqrt{\tfrac{s_i}{W}} - \sqrt{\tfrac{s^{B}_i}{W_B}}\right)^{2}}, \qquad W = \sum_i s_i
```

Spreading.

- **Model.** SI, as in the contact-network literature. A seed starts infected, and at each step an infected node infects each susceptible neighbour j with probability min(1, beta times p_ij). Backbones keep the original p_ij on retained edges. Beta is set per network so that the median half-infection time on G is about 20 steps, then held fixed for every backbone. Runs stop at ten times that median, and nodes never infected get that time.
- **Seeds.** Following the protocol of Pereira et al. (2026): 15% of nodes, namely the 5% most, 5% median and 5% least central by eigenvector centrality in G, with 100 runs per seed.
- **F_order.** For each seed, the Spearman correlation between the mean infection times of the nodes in G and in B, clipped at 0, averaged over seeds.
- **F_time.** For each seed, the ratio of median half-infection times in G and in B, clipped to [0, 1], averaged over seeds.
- **F_spread** is the mean of F_order and F_time.
- **Robustness check F_SIR.** Reed-Frost SIR with one random seed per run and 200 runs, at ten transmission levels from no outbreak to saturation on G. F_SIR is one minus the mean absolute difference between the mean final outbreak fractions on G and on B.
- **Variance reduction.** Runs on G and B share random numbers keyed by run, step and edge, so differences reflect the backbone and not simulation noise.

Communities. Louvain with weights, resolution 1 and 20 random seeds, on G and on B. An isolated node in B is a singleton community. Louvain disagrees with itself, so the score is normalised by that ceiling:

```
F_{\mathrm{comm}} = \min\!\left(1, \frac{\overline{\mathrm{NMI}}(P_G, P_B)}
{\overline{\mathrm{NMI}}(P_G, P_G')}\right)
```

The numerator averages NMI over all pairs of runs on G and B. The denominator averages NMI over pairs of different runs on G. On synthetic networks, also report the adjusted Rand index of G and of B against the planted partition, because preserving the communities found in G and recovering the true communities are different questions.

Paths. Let D_G and D_B be shortest-path distances under d_ij, and P the set of node pairs connected in G. Because B is a subgraph of G, D_B is never smaller than D_G, and a pair disconnected in B counts as a failure:

```
F_{\mathrm{path}}(\varepsilon) = \frac{\left|\{(u,v)\in\mathcal{P} : D_B(u,v)
\le (1+\varepsilon)\,D_G(u,v)\}\right|}{|\mathcal{P}|}
```

The default is epsilon = 0.1, varied in E5. Also report the median stretch D_B / D_G among connected pairs and the share of disconnected pairs. For networks beyond a few hundred nodes, sample node pairs.

Rankings. For each centrality c (strength, eigenvector centrality of the proximity matrix, and betweenness under d_ij), compute the Spearman correlation between c on G and c on B, with average ranks for ties, so isolates tie at the bottom:

```
F_{\mathrm{rank},c} = \max\{0,\ \rho_{\mathrm{Spearman}}(c_G, c_B)\}, \qquad
F_{\mathrm{rank}} = \tfrac{1}{3}\sum_c F_{\mathrm{rank},c}
```

Top-10% overlap is a secondary score.

Baselines and skill score. For each function and budget, compute the fidelity of each null baseline averaged over 20 draws. The skill score rescales a method's fidelity against chance:

```
S_{m,g} = \frac{F_{m,g} - \bar F_{\mathrm{rand},g}}{1 - \bar
F_{\mathrm{rand},g}}
```

It is defined only when the denominator is at least 0.05. Otherwise report the plain difference and mark the case as near ceiling. A negative score means worse than random removal. Report the score against both nulls: R-W, which is weight-aware, is the more demanding test.

Uncertainty

Source	Where it arises	How it is handled
Simulation noise	SI and SIR runs, seed choice	Bootstrap over runs and seeds with 1,000 resamples and 95% percentile intervals; paired bootstrap on the same resamples for differences between methods
Algorithmic randomness	Louvain seeds, tie-breaking, random baselines	Average over 20 Louvain seeds and 20 baseline draws, and resample them in the bootstrap

Source	Where it arises	How it is handled
Network variation	Only a few real networks; synthetic instances	Synthetic ensembles with at least 10 instances per setting; cluster bootstrap over networks and mixed-effects models with network as a random effect; always print the number of independent networks next to a claim

Scores that are deterministic (structure, paths, rankings) have no simulation noise, only tie-breaking variation.

Structure-to-function gap and mechanism measures. The gap is F_struct minus F_g, for each method, network, budget and function, and a fidelity failure is defined in section 4. The mechanism measures, computed per backbone, are the bridge-retention ratio and the hub concentration:

```
\rho_{\mathrm{br}} = \frac{|E_B^{\mathrm{inter}}| \,/\, |E^{\mathrm{inter}}|}
{|E_B^{\mathrm{intra}}| \,/\, |E^{\mathrm{intra}}|}
```

Inter and intra refer to the planted partition on synthetic networks and to the Louvain partition of G on real ones. Hub concentration is the share of retained weight on edges that touch the top 10% of nodes by strength. Network descriptors for the regressions are density, coefficient of variation of weights, the Spearman correlation between edge weight and the product of endpoint degrees (weight-degree coupling), clustering, modularity of the Louvain partition, the share of semi-metric edges (one minus metric-backbone size over edge count) and size.

7. Recommendation engine

The Backbone Advisor recommends a method for a given network, objective and edge budget, and shows the evidence and trade-offs behind the recommendation.

Input	Options
Network	A weighted edge list, or a preset network
Objective	One function, or weights υ over spreading, communities, paths and rankings that sum to 1

Input	Options
Edge budget	A share q, or a range
Constraint	Optional: the backbone must be connected

Evidence mode (the primary path). It computes the answer on the user's own network instead of predicting it.

1. Convert weights to proximity and distance, and compute the network descriptors.
2. Run the six methods at budget q and score each backbone on the four functions, with the null baselines. Reuse cached results where they exist.
3. Compute the utility U_m, the sum over functions of u_g times F_m,g, with a 95% interval and the probability that m is best, from a paired bootstrap over runs and seeds.
4. Recommend the method with the highest mean utility. Label the recommendation clear when its probability of being best is at least 0.8, and a close call when the intervals of the top two overlap, showing both. If the connected-backbone constraint is on, exclude methods whose backbone at q is disconnected.
5. Show the trade-offs: the set of methods not dominated on the four function scores, and what the recommended method gives up on each function against the best method for that function.
6. Show the evidence: each function's fidelity with its interval, the skill score against both nulls, the structure-to-function gap with a failure warning where it applies, mechanism readouts (bridge-retention ratio, hub concentration, connectivity floor) and a sensitivity check of whether the recommendation changes one budget step either way or under the alternative proximity map.
7. Export the decision, its settings and its seeds as a report.

Predictive mode (conditional). Its purpose is to answer without running all six methods, for example on a network too large for evidence mode. It maps cheap descriptors of the network, the budget and the objective to the expected fidelity of each method, using gradient-boosted trees or ridge regression, and attaches a conformal prediction interval. It flags low confidence when the intervals of the top two methods overlap or when the network lies outside the descriptor range seen in training. It is included only if it passes the validation below.

Validation	What it tests
Leave one network out	An unseen network from a seen domain
Leave one domain out (contact, transport, other)	An unseen domain
Train on synthetic, test on real	Transfer from controlled to real networks
Locked test set of three real networks, untouched until E6	The final unbiased estimate

Baselines are a random method, the single method that is best on average (global best), the literature rule of always choosing SMDS, and the nearest training network by descriptors. Metrics are top-1 accuracy, regret (utility of the best method minus utility of the chosen one, as a mean and a 90th percentile), rank correlation between predicted and true method order, and interval coverage against nominal. Pass criteria: lower regret than the global-best baseline with a paired interval that excludes zero, and coverage within five points of nominal. Otherwise drop predictive mode, keep evidence mode and report the negative result.

Limits. With few real networks, any claim of generalization is limited to the tested domains and size range. Evidence mode is exact for the given network and dynamics, so it is the primary recommendation path. For networks larger than the main-experiment size, reduce the seeds and sample node pairs, and say so in the report.

8. Experimental design

The design tests H1 to H5 (section 4) and validates the advisor (section 7). Metrics, thresholds and tests are fixed in a protocol file before E1 to E6 run, and any later change is labelled exploratory.

Experiment	Question	What varies	Output
E1 Agreement	Does structural fidelity rank methods like functional fidelity does? (H1)	Method, network, budget	Kendall's tau per cell, its mean with a cluster-bootstrap interval, pooled regression

Experiment	Question	What varies	Output
E2 Failure atlas	Where do failures occur? (RQ1, RQ3)	Method, budget, network, function	Failure-rate maps and gap heatmaps
E3 Mechanisms	Why do they occur? (H2, H3)	Real networks: regression on mechanism measures. Synthetic: a factorial design	Effect sizes for bridge retention, hub concentration, weight heterogeneity, weight-degree coupling and community mixing
E4 Method contrasts	Which method wins where? (H4, H5)	Method, budget	Skill scores against both nulls, paired differences, winners per cell
E5 Robustness	Do the verdicts survive other modelling choices?	SI versus SIR, beta calibration, proximity map, epsilon, Louvain resolution, seed protocol	Whether each verdict changes
E6 Advisor validation	Does the advisor generalize? (RQ5)	Evidence-mode checks; predictive-mode splits	Regret, accuracy, interval coverage

Networks. Confirm each download in week 1.

Network	Type	Size	Source
Hospital, workplace, conference, second workplace, primary school, high school	Face-to-face contact	75 to 327 nodes; 755 to 8,317 edges (as reported in Table 1 of Pereira et al. (2026), which also lists the original studies)	SocioPatterns and the studies cited there
US air transportation	Transport	About 380 nodes and 9,700 edges; passengers	The Serrano et al. network, published with NetBone

Network	Type	Size	Source
Les Miserables	Co-occurrence, sanity check	77 nodes, 254 edges	Built into NetworkX
At least three more	Other domains (transport, biological, economic)	Up to about 1,000 nodes and 50,000 edges	Weighted networks from the Netzschleuder repository, which Kirkley (2025) used; check weights and licences
Synthetic	Controlled	300 nodes	Weighted community model, below

The contact networks share one domain, so the other rows are what let you say anything about generalization.

Synthetic networks for causal tests. A weighted community model with 300 nodes and three factors: community mixing (0.1, 0.3, 0.5), log-normal weight spread (sigma 0.5, 1.0, 1.5) and weight-degree coupling (0, 0.5, 1; the weight is multiplied by the product of endpoint degrees raised to this power, then rescaled to keep the mean weight fixed). Ten instances per cell give 270 networks. A separate pilot set, not part of the 270, is used only to set tau_s, tau_f and beta. Lock three real networks from different domains as the test set for E6.

Compute plan (to be profiled in week 2).

Item	Count
Real networks	About 12
Method backbones per real network	6 methods times 8 budgets = 48
Null backbones per real network	2 nulls times 8 budgets times 20 draws = 320
SI runs per method backbone	15% of nodes as seeds times 100 runs (about 3,800 for 250 nodes)
SI runs per null backbone	The same seeds times 20 runs (about 760)

Item	Count
Synthetic networks	270, with a reduced protocol of 9 seeds times 50 runs per method backbone, and 10 null draws with 9 seeds times 20 runs
Total SI runs	About 19 million

At about 1 ms per run with compiled code, that is roughly five CPU-hours. If profiling shows more, reduce seeds, runs or null draws first, and keep the budget grid.

Reproducibility

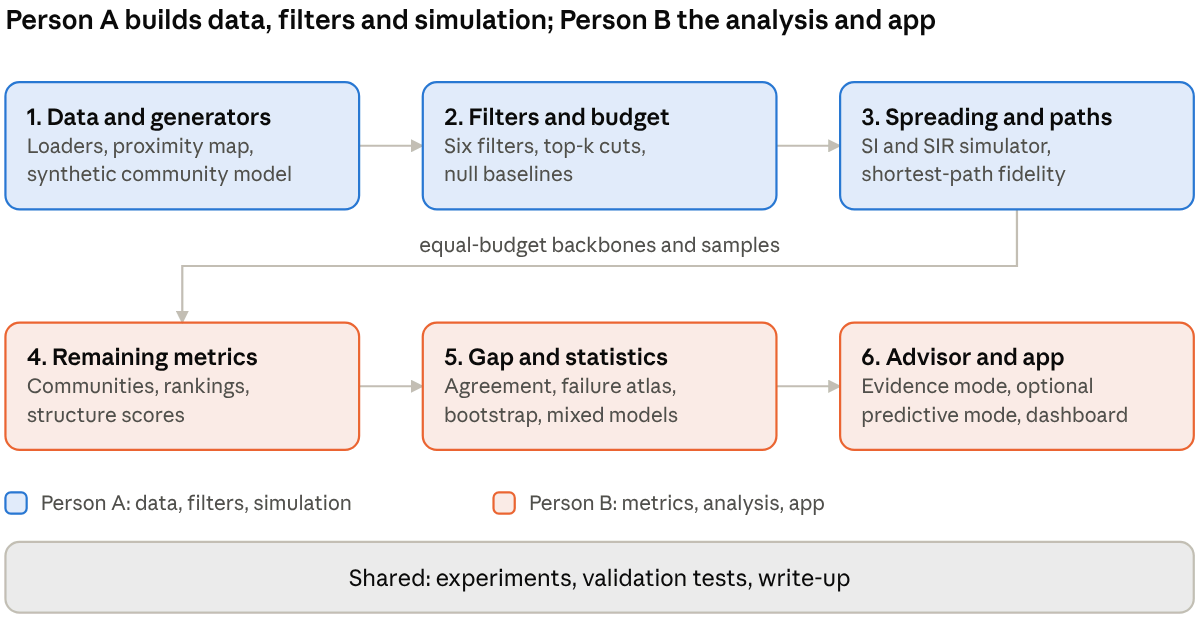
- Freeze hypotheses, thresholds, metrics and tests in a protocol file and tag it in version control before running E1 to E6.
- One global seed, with a derived seed for each component, all recorded in configuration files. Every figure regenerates with one command from stored results.
- Record each dataset's source, access date and checksum, and pin the software environment with a lock file.
- Unit tests: every score equals 1 at q = 1; SMDS at the metric-backbone size has F_path = 1 and preserves all shortest paths; scores stay in [0, 1]; scores are invariant to node relabelling; random-baseline fidelity rises with q; effective resistance is correct on a path and a triangle; DF, NC and HSS agree with NetBone on Les Miserables.

Conclusions the design cannot support

- Causal statements from regressions on real networks. Use the synthetic factorial experiments for causal language.
- Claims beyond the tested domains and size range, or beyond SI and SIR with this weight-to-probability map.
- Any claim of novelty beyond the assessment in section 3.

9. Implementation architecture and work split

Six modules form one pipeline. Person A owns the first three: data and generators, the six filters with the budget controller, and the spreading and path metrics. Person B owns the last three: community, ranking and structure metrics, the gap analysis and statistics, and the advisor with the dashboard. Both share the experiments and the validation tests.



architecture and work split · 6 modules, 2 owners

Module 2 hands every backbone onward at an equal budget, which is what makes the comparison fair. The metric modules return each score together with the samples behind it, so module 5 can bootstrap them. The dashboard reads results from every stage. The interface contract is simple: a backbone is a ranked edge list plus a budget, and a metric returns a score plus its samples.

Suggested repository layout

```
backbone-bench/
  data/          # loaders, proximity transform, synthetic generator
  filters/       # threshold.py, disparity.py, noise_corrected.py,
                 #   smds.py, hss.py, eff_resistance.py
  budget/        # budget.py (top-k by score, ties, floors), nulls.py (R-U, R-
W)
  metrics/       # structure.py, spreading.py (SI, SIR), paths.py,
                 #   communities.py, ranks.py
  analysis/      # skill.py, bootstrap.py, gap.py, mechanisms.py,
mixed_models.py
  advisor/       # evidence.py, predictive.py (conditional), report.py
  experiments/   # e1_agreement.py ... e6_advisor.py, protocol.md (frozen)
  app/           # streamlit_app.py and pages
  tests/         # unit tests and NetBone agreement checks
```

Tools. Python with NumPy and SciPy (sparse matrices), Numba or another compiled kernel for the SI and SIR loops, NetworkX for graph containers and Louvain, statsmodels for the mixed-effects models, scikit-learn for predictive mode, and NetBone only to validate. Write the filters, the budget controller and the epidemic simulator yourselves. Everything runs on a CPU: parallelize runs across cores and store results as parquet files so figures regenerate without recomputing.

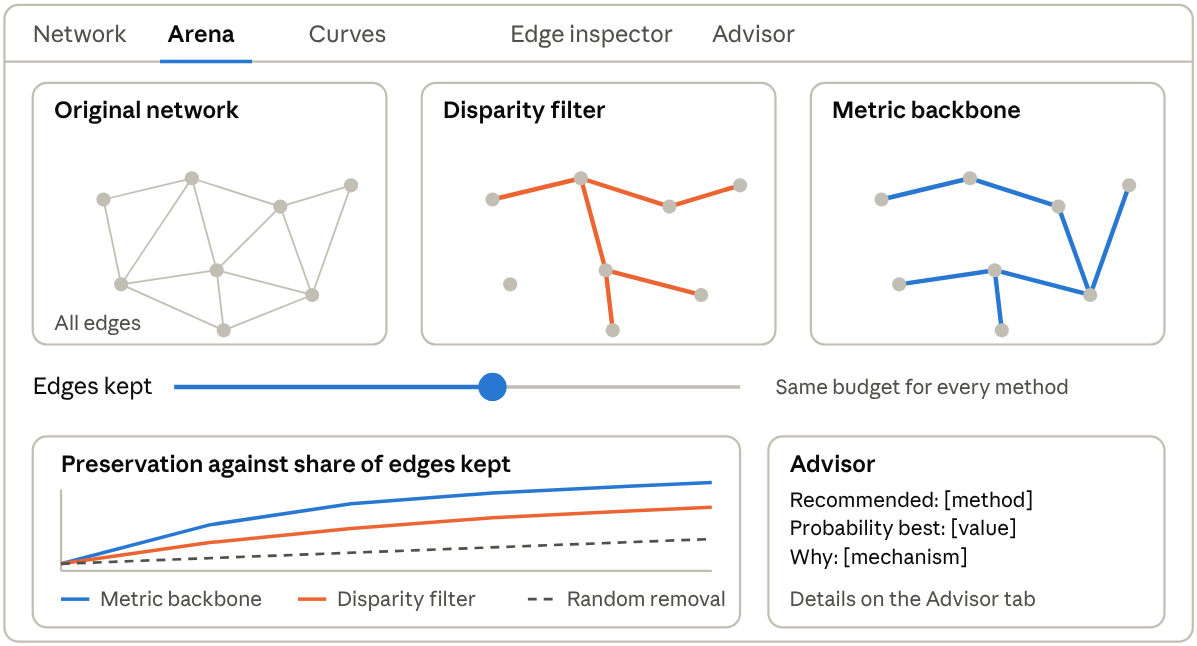
10. Demo and dashboard

The dashboard lets an audience watch backbones shrink a network, see which ones still behave like the original, find the cases where structural checks misled, and get a recommendation with its evidence. It has six linked components, and the budget slider, the selected edge and the objective weights are shared between them.

Component	What it does	Question it answers
Network view	An interactive drawing of the original or any backbone. Edge width shows weight, node size shows strength, colour shows community. Zoom, hover and click an edge	What does each method keep?
Method arena	The original and the six backbones at one budget in a grid, each with score bars. Pick two methods for a head-to-head overlay that colours edges kept by both, by one or by neither	How do the methods differ at the same budget?
Preservation curves	Fidelity against budget for each function, with the six methods and both nulls, 95% bands and a budget cursor linked to the arena	Where does each method hold up, and where does it fall below chance?
Failure atlas	A heatmap of the structure-to-function gap over methods and budgets, with failures outlined. Click a cell to load that case into the arena	When does a structural check mislead? (H1, E2)

Component	What it does	Question it answers
Edge inspector	Click an edge to see its score and rank under every method, whether each keeps it, and its role: bridge between communities, on a shortest path, or touching a hub	Why was this edge kept or dropped? (H2)
Advisor panel	Objective weights, budget and constraint in. Recommendation, probability of being best, trade-offs, evidence and warnings out, with a report export	Which method should I use, and how sure is that?

Arena screen: every method at the same edge budget, side by side

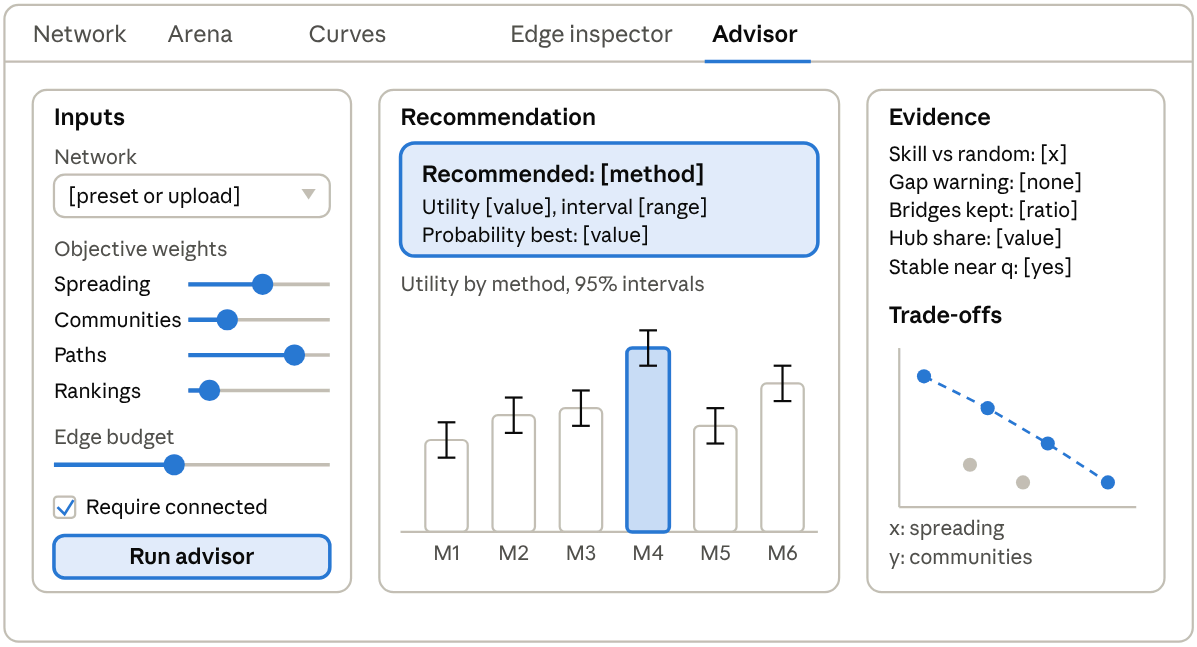


UI wireframe · Arena screen

The arena wireframe above is the main screen. Every method is cut to the same edge budget, so any difference the audience sees comes from the method and not from keeping more edges.

The advisor screen puts the explainable recommendation on one page: inputs on the left, the recommendation with its uncertainty in the centre, and the evidence and trade-offs on the right. The bracketed items are filled from real runs.

Advisor screen: a recommendation with its evidence, uncertainty and trade-offs



UI wireframe · Advisor screen

The bars show the utility of each method with its interval, so a close call is visible as overlapping whiskers. The trade-off panel shows which methods are not dominated across the functions.

Suggested stack and performance. A Python backend with your own filters and simulator, Streamlit for the pages and Plotly for the network views, curves and heatmap. Plotly handles a few hundred nodes comfortably with a force layout cached per network. Precompute backbones and scores for the preset networks so the dashboard responds instantly, and compute live only for uploaded networks, with reduced seeds and a progress indicator. Keep the shared controls in one state object.

A five-minute demo script

1. Load a contact network and show the dense original in the network view.
2. Move the budget slider. Six backbones appear in the arena, each keeping visibly different edges.
3. Open the failure atlas and pick a cell where the structural check looked fine but function broke. Load it into the arena.
4. Open the edge inspector on a bridge edge: which methods dropped it, and the spreading curve that follows. Show this only if E3 supports the bridge mechanism. Otherwise show what the results do support.

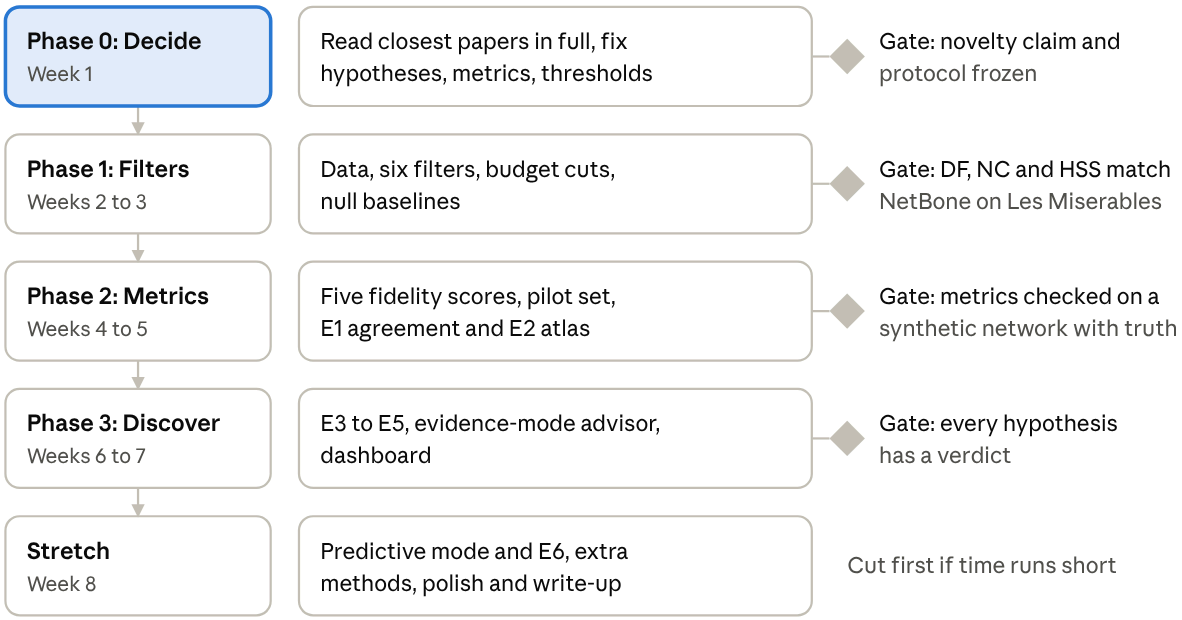
5. Open the curves to show the same failure against the random baselines.
6. End on the advisor: set objective weights, read the recommendation with its probability of being best, its trade-offs and any warning.

Build the script from the results. If a hypothesis was rejected, show that case honestly.

11. Timeline, feasibility and risks

The plan assumes eight weeks and is built so that stopping after any gate still leaves a defensible project. If your deadline is shorter, compress the phases in proportion and keep the week-1 decision.

Eight weeks, four gates, and a week-1 decision that shapes everything



roadmap · 5 phases, 4 gates

Phase 0 matters most. It fixes the novelty claim after you have read the closest papers in full (section 3), so everything after it is aimed at a claim that survives them.

Feasibility tiers. Finish each tier before starting the next.

Tier	Contents	Rule
P0, core	Six filters, budget controller and null baselines, structure, spreading, community, path and ranking metrics, E1 and E2, the advisor in evidence mode, a basic dashboard	Must be finished
P1, discovery	E3 mechanism tests with the synthetic factorial design, E4 method contrasts, E5 robustness, the failure atlas and trade-off views	Start once P0 works, around week 5
P2, optional	Predictive mode and E6 generalization, extra methods (LANS, Distance Backbone Synthesis), larger networks, a map view	Cut first if time runs short

CPU-friendly by design. Main experiments use networks up to a few hundred nodes and tens of thousands of edges, compiled kernels for the epidemic loops, cached results, reduced protocols for the synthetic sweeps and the run counts in section 8. Profile in week 2 and adjust seeds, runs and null draws before changing the budget grid or the method set.

Risks

Risk	Why it matters	Fallback
The closest papers already answer the central question	The novelty claim fails	Decide in week 1, after reading Hamann, Yan, Correia and Soriano-Paños in full, and narrow the project to the part that remains, for example the mechanism tests on synthetic networks
Failures are rare, or H1 is rejected	There is no discovery to report	Report it: structure predicts function here, and the advisor can then rely on structural proxies, which is a useful result

Risk	Why it matters	Fallback
Too few real networks	Cross-network claims are weak	Synthetic ensembles with known truth, a larger real corpus from Netzschleuder, and network counts printed beside every claim
Simulation cost exceeds the CPU budget	Experiments do not finish	Profile in week 2, then reduce seeds, runs and null draws, sample node pairs and keep to the size limits
Errors in the filters	Wrong conclusions	NetBone agreement for DF, NC and HSS, independent checks for SMDS and wEffRes, and unit tests
Predictive mode does not generalize	Overclaiming	Pass criteria fixed in advance; otherwise drop it and keep evidence mode
Scope creep	The core stays unfinished	Optional items are cut first
A dataset cannot be downloaded	An experiment loses a network	Confirm access in week 1 and fall back to the NetBone example data and synthetic networks

Planned paper structure

1. Introduction: backbones, structural checks and the structure-versus-function question.
2. Related work and the remaining gap (section 3).
3. Methods: six filters, matched budgets and fidelity scores (sections 5 and 6).
4. Results: agreement and the failure atlas, mechanisms, method contrasts and robustness.
5. The advisor and its validation.
6. Discussion: when structural checks can be trusted and when they cannot.
7. Limitations: few real networks, static aggregation of contact data, SI and SIR with one weight-to-probability map, and the size range.

8. Reproducibility: protocol, seeds, code and dataset checksums.

12. References

I read the full texts of references 11, 12 and 13 (Pereira et al., Kirkley, NetBone) and abstracts or summaries of the others. Check each full text before citing it.

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